Ablation studies

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete ablation condition and failure record.

Keep supplied counts, controls, missing cells, severity and scope together.

Keep one habit: Recall → W1–W3/W5 → independent fifteen-case comparison → research project → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Multiple comparisons. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Effect size and practical importance. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Confounders and controls. Give an example and a mistake that the idea helps you avoid.

Multiple comparisons

My remembered idea, example and correction

Effect size and practical importance

My remembered idea, example and correction

Confounders and controls

My remembered idea, example and correction

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- An ablation removes a component while preserving evaluation cases, scoring and other settings. Main full system has retrieval and validation with 8/10 passes. Without retrieval it has 5/10, a net drop of three passes or thirty percentage points. Without validation it has 7/10, a drop of one pass or ten points, and a stipulated unauthorized action. The action boundary is not captured by score magnitude alone.

  Joint removal is not supplied. If its count were hypothetically 4, the interaction contrast 8 – 7 – 5 + 4 would be zero; if hypothetically 6, it would be two. Neither value is observed. Thus the supplied three cells do not determine interaction or additive component contributions.

  Separate filter example has 8/10 full versus 6/10 without the filter: a net two-pass, twenty-percentage-point drop, or 25% relative decrease. Net counts do not identify exact flipped cases. Also changing the model confounds filter-only attribution. These supplied summaries establish no actual model execution, significance or universal causal effect.

  Preserve conditions and inspect per-case failure categories when records exist.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

- An ablation removes a component while preserving evaluation cases, scoring and other settings. Main full system has retrieval and validation with 8/10 passes. Without retrieval it has 5/10, a net drop of three passes or thirty percentage points. Without validation it has 7/10, a drop of one pass or ten points, and a stipulated unauthorized action. The action boundary is not captured by score magnitude alone.

  Joint removal is not supplied. If its count were hypothetically 4, the interaction contrast 8 – 7 – 5 + 4 would be zero; if hypothetically 6, it would be two. Neither value is observed. Thus the supplied three cells do not determine interaction or additive component contributions.

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  Preserve conditions and inspect per-case failure categories when records exist.

- Full system 8/10, without retrieval 5/10, without validation 7/10; the last permits an unauthorised action. Show the intermediate steps and give the conclusion with its scope.

Condition	Pass count/rate	Net drop from full	Interpretation
Full system	8/10 = 80%	Reference	Both components present
Without retrieval	5/10 = 50%	3 passes; 30 percentage points	Conditional removal effect in this configuration
Without validation	7/10 = 70%	1 pass; 10 percentage points	Also permits a stipulated unauthorized action
Without both	Unknown	Unknown	Cannot calculate joint effect or interaction

Condition	Pass count/rate	Net drop from full	Interpretation
Changed case identities	Unknown	Net count is not a flip ledger	Inspect per-case records if collected

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- An ablation removes a component while preserving evaluation cases, scoring and other settings. Main full system has retrieval and validation with 8/10 passes. Without retrieval it has 5/10, a net drop of three passes or thirty percentage points. Without validation it has 7/10, a drop of one pass or ten points, and a stipulated unauthorized action. The action boundary is not captured by score magnitude alone.

  Joint removal is not supplied. If its count were hypothetically 4, the interaction contrast 8 – 7 – 5 + 4 would be zero; if hypothetically 6, it would be two. Neither value is observed. Thus the supplied three cells do not determine interaction or additive component contributions.

  Separate filter example has 8/10 full versus 6/10 without the filter: a net two-pass, twenty-percentage-point drop, or 25% relative decrease. Net counts do not identify exact flipped cases. Also changing the model confounds filter-only attribution. These supplied summaries establish no actual model execution, significance or universal causal effect.

  Preserve conditions and inspect per-case failure categories when records exist.

- A report omits this qualification: Removing a component changes a complete procedure; attribution depends on the controlled design and possible **interaction**s.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- An ablation removes a component while preserving evaluation cases, scoring and other settings. Main full system has retrieval and validation with 8/10 passes. Without retrieval it has 5/10, a net drop of three passes or thirty percentage points. Without validation it has 7/10, a drop of one pass or ten points, and a stipulated unauthorized action. The action boundary is not captured by score magnitude alone.

  Joint removal is not supplied. If its count were hypothetically 4, the interaction contrast 8 – 7 – 5 + 4 would be zero; if hypothetically 6, it would be two. Neither value is observed. Thus the supplied three cells do not determine interaction or additive component contributions.

  Separate filter example has 8/10 full versus 6/10 without the filter: a net two-pass, twenty-percentage-point drop, or 25% relative decrease. Net counts do not identify exact flipped cases. Also changing the model confounds filter-only attribution. These supplied summaries establish no actual model execution, significance or universal causal effect.

  Preserve conditions and inspect per-case failure categories when records exist.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Full 12/15, without A 9/15, without B 11/15. Compute sample drops and state **interaction** limit.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- Add a labelled **ablation** studies **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **ablation** studies.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for **Ablation** studies. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Ablation** studies. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Full 12/15, without A 9/15, without B 11/15. Compute sample drops and state **interaction** limit.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Remove one component at a time. A retrieval pipeline has search plus a relevance filter. A team tests the filter's contribution.

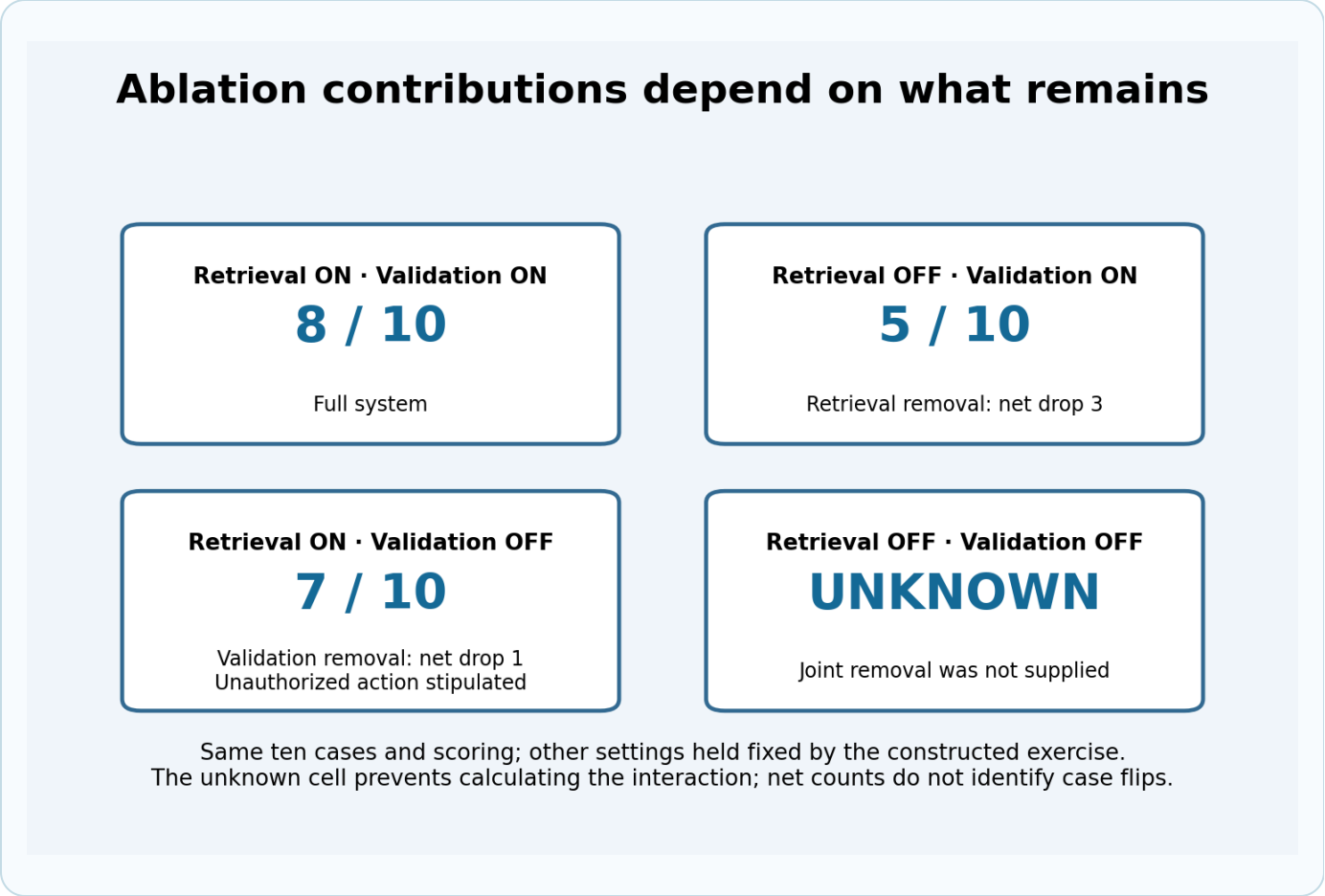
- Full pipeline: 8/10 correct. Filter removed: 6/10 on the same cases, with all other settings fixed.

- Calculate the filter **ablation**'s two rates and count change. Which single component is altered, and why does also changing the model prevent isolating the filter?

- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

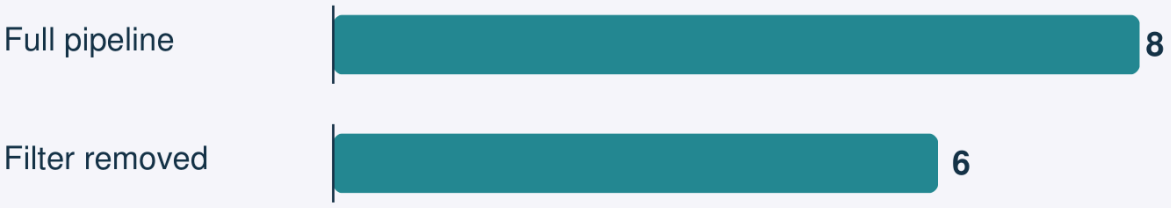
  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.



AI ACADEMY / AI-7/16

Relevance-filter ablation

correct answers out of 10 shared cases



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/16

Ablation studies

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Name the component

The relevance filter is the single declared change.



2. Run the ablation

Remove only the filter and preserve cases and scoring.



3. Compare outcomes

Correctness falls from 80% to 60%, two cases in this set.

CHANGE ONE THING

Remove the filter and also change the model version.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Remove one component at a time. A retrieval pipeline has search plus a relevance filter. A team tests the filter's contribution.
- Which labelled step matches this detail? Correctness falls from **80%** to **60%**, two cases in this set.
- 1. Name the component
- 2. Run the **ablation**
- 3. Compare outcomes
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Compare outcomes Correctness falls from **80%** to **60%**, two cases in this set. An **ablation** probes a component by removing or changing it while holding the rest fixed.

It supports a scoped comparison, not a claim that the component always helps.

Changed-case question and optional hint

- Changed-case practice: Remove the filter and also change the model version. Which conclusion is supported by the supplied facts?
- 1. The comparison no longer isolates the filter.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The comparison no longer isolates the filter. Two simultaneous changes provide alternative explanations.

Repair and retain the record

Review Multiple comparisons (ai-7/15). Goal: Explain the **mechanism** and terminology of multiple comparisons. Short practice: Calculate the six-test cutoff and identify which supplied p-values reject.

Why must the team count every planned comparison rather than report only its smallest p-value?. Return to this week and explain what you changed.

Review Effect size and practical importance (ai-7/13). Goal: Explain the **mechanism** and terminology of effect size and practical importance. Short practice: Calculate the absolute error-rate reduction from 20/1000 to 15/1000.

Does it meet the ten-fewer-errors threshold, and what changes when B has eight errors?. Return to this week and explain what you changed.

Review Confounders and controls (ai-7/10). Goal: Explain the **mechanism** and terminology of confounders and controls. Short practice: Draw the four condition/background combinations.

Which variables were coupled before, and why would collecting only healthy-white and spotted-blue again leave the confounding unresolved?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Ablation studies: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

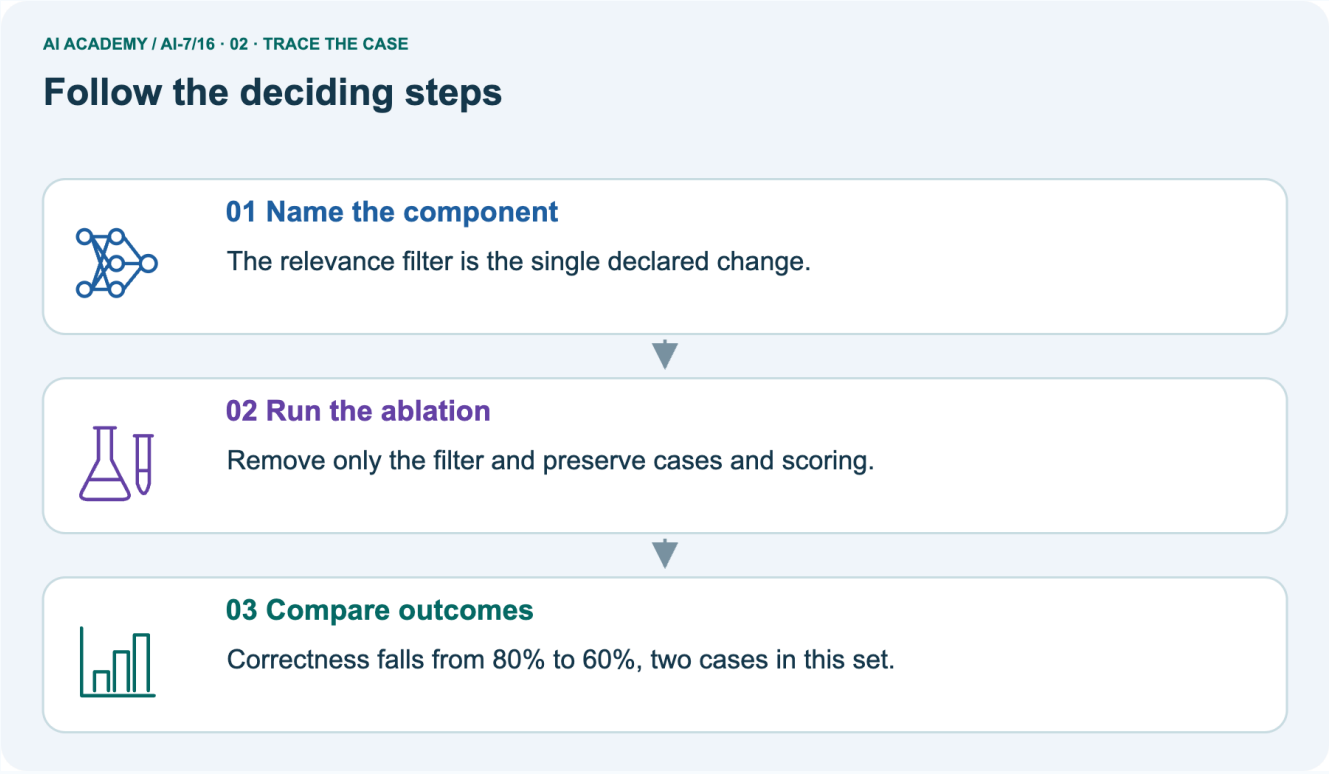
Remove one component at a time. A retrieval pipeline has search plus a relevance filter. A team tests the filter's contribution. Supplied facts: Full pipeline: 8/10 correct.

Filter removed: 6/10 on the same cases, with all other settings fixed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



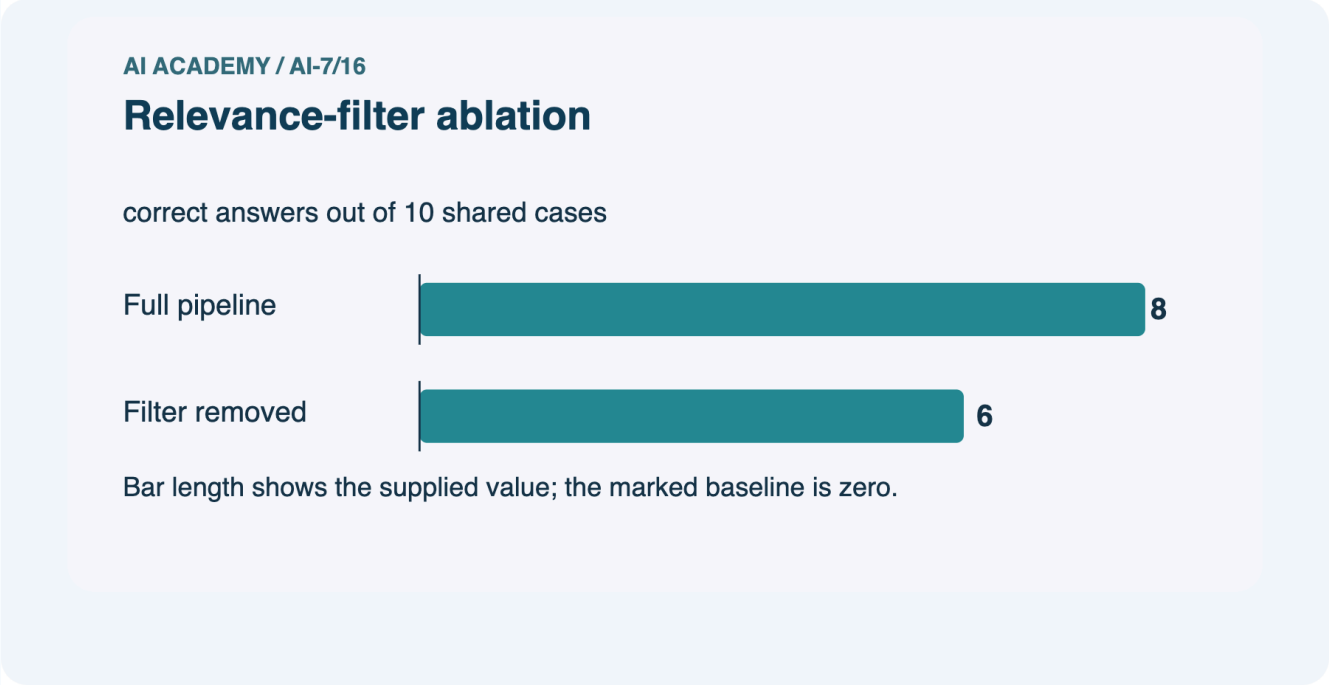
Follow the labelled picture with the matching explanation. Open the vector diagram

Name the component: The relevance filter is the single declared change. → Run the **ablation**: Remove only the filter and preserve cases and scoring. → Compare outcomes: Correctness falls from **80%** to **60%**, two cases in this set.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Relevance-filter ablation



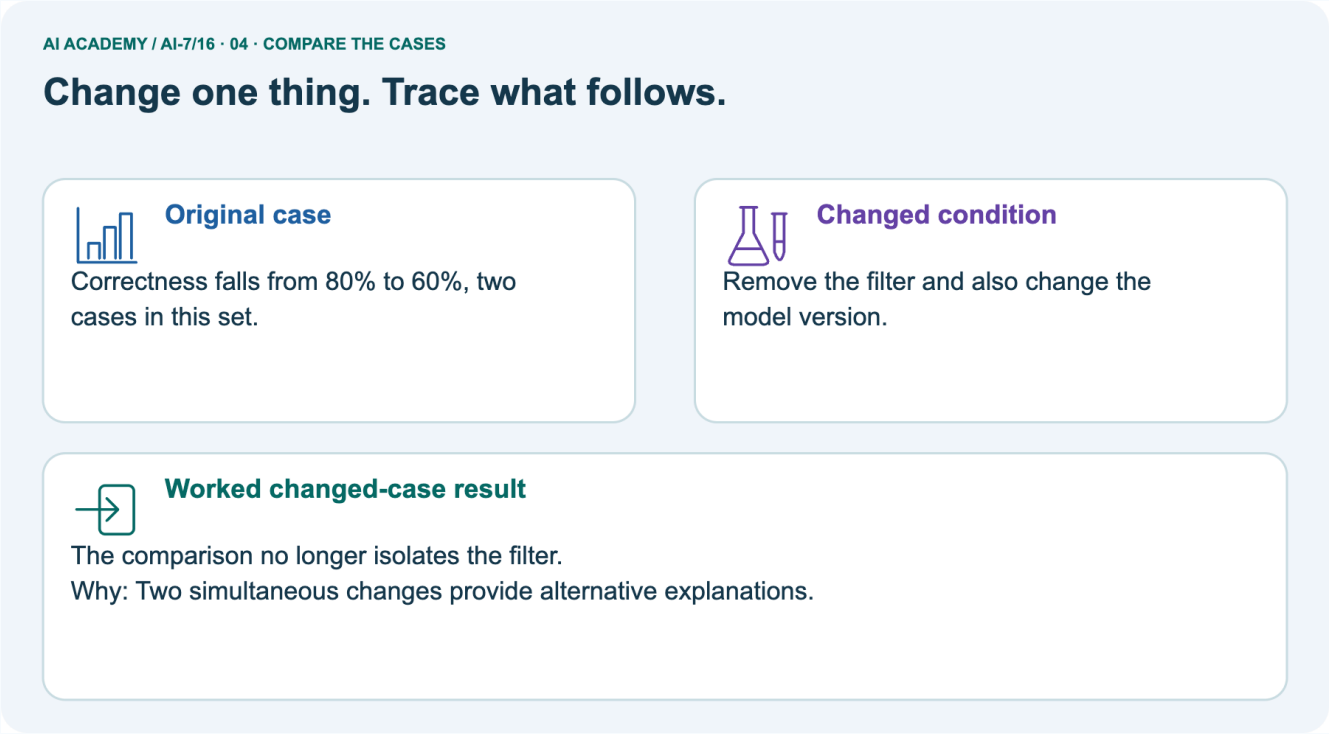
Follow the labelled picture with the matching explanation. Open the vector diagram

Relevance-filter **ablation**. Units: correct answers out of 10 shared cases. Full pipeline: 8. Filter removed: 6.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

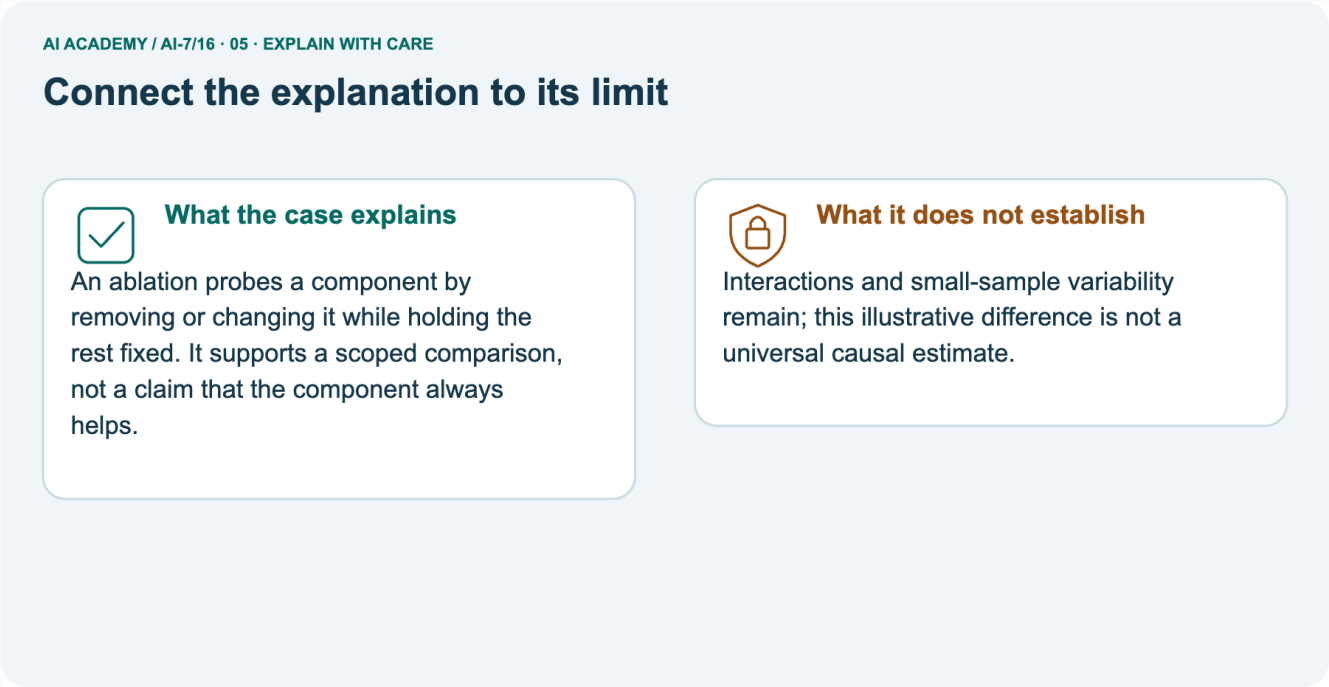
Original case: Correctness falls from **80%** to **60%**, two cases in this set. Changed condition: Remove the filter and also change the model version. Expected result: The comparison no longer isolates the filter.

Reason: Two simultaneous changes provide alternative explanations.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

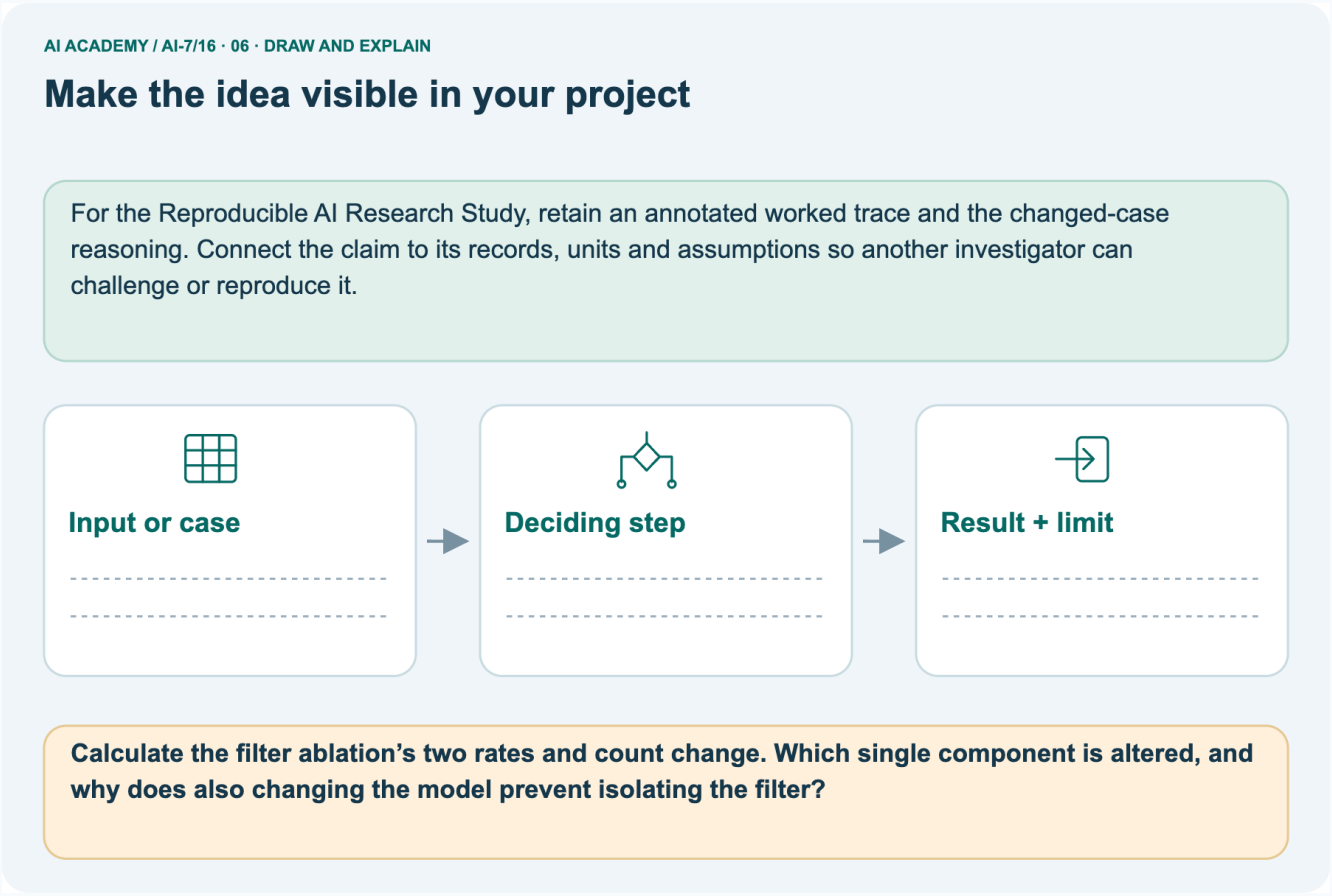
An **ablation** probes a component by removing or changing it while holding the rest fixed. It supports a scoped comparison, not a claim that the component always helps.

Limit: **Interaction**s and small-sample variability remain; this illustrative difference is not a universal causal estimate.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the filter **ablation**'s two rates and count change.

Which single component is altered, and why does also changing the model prevent isolating the filter?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Ablation studies: the case

02. Follow the deciding steps

03. Relevance-filter ablation

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.